

Data Dictionary

PERSON

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique person identifier	INT	PK	NOT NULL, UNIQUE	103 (generated)
first_name	Person first name	VARCHAR(50)		NOT NULL	John
last_name	Person last name	VARCHAR(50)		NOT NULL	Smith
email	Patron contact email / login	VARCHAR(255)		NOT NULL, UNIQUE, CONTAINS '@'	John.smith@email.com
phone	Person contact phone number	VARCHAR(20)		NOT NULL, UNIQUE	1+4035556767888
password	Person hashed password	VARCHAR(255)		NOT NULL	(hashed string)

PERSON: One person is one human in the real world, either patron or staff or both.

PATRON

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Person id	INT	PK / FK	NOT NULL, UNIQUE	103 (from person)
role_id	Patron's role identifier	INT	FK	NOT NULL (References ROLE.id)	3 (from role)

PATRON: One person has one patron account to buy tickets with.

STAFF

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Person id	INT	PK / FK	NOT NULL, UNIQUE	103 (from person)
role_id	Staff members role identifier	INT	FK	NOT NULL (References ROLE.id)	1 (from role)

STAFF: One person can have one Staff account for managing events.

VENUE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique venue identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
name	Venue's name	VARCHAR(50)		NOT NULL	Playwright West
address	Street address of the venue	VARCHAR(255)		NOT NULL	123 Theater Way
city	City the venue is located in	VARCHAR(100)		NOT NULL	Calgary
province	Province the venue is located in	VARCHAR(50)		NOT NULL	Alberta
postal_code	Postal code of the venue	VARCHAR(20)		NOT NULL	T2P 1J9

VENUE: one row is one physical building in real life.

EVENT

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique event identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
title	Event title	VARCHAR(100)		NOT NULL	Same Songs New Play
description	Event description	VARCHAR(500)			Play about writing new songs for a new play
status	Status of the event	VARCHAR(50)		NOT NULL	'DRAFT' (default)
live_date	Date and time the event goes live	TIMESTAMPZ			2026-11-01T09:00:00Z
image	Promotional image for the event	BYTEA			(binary blob data)

EVENT: one row is one event that is happening.

STAGE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique stage identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
venue_id	Venue identifier stage belongs to	INT	FK	NOT NULL	12 (from venue)
name	Stage's name	VARCHAR(50)		NOT NULL	Main Stage

STAGE: one row is one stage that is inside a venue.

PERFORMANCE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique performance identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
event_id	Event identifier performance belongs to	INT	FK	NOT NULL	12 (from event)
stage_id	Stage identifier performance belongs to	INT	FK	NOT NULL	12 (from stage)
date_time	Date and time of performance	TIMESTAMP		NOT NULL	2026-10-22T13:30:00
filled	Status if all seats are filled for performance	BOOLEAN		NOT NULL	FALSE (default)

PERFORMANCE: One row is one performance for a event.

SEAT

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique seat identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
row	Row seat lives at	VARCHAR(5)		NOT NULL	A
seat_number	Number of seat in row	INT		NOT NULL	15
section_id	Section seat lives in	INT	FK	NOT NULL (References SECTION.id)	4 (from section)

SEAT: one row is one physical seat belonging to one stage in real life.

PERFORMANCE SEAT

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique event seating identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
seat_id	Seat identifier	INT	FK	NOT NULL	12 (from seat)
performance_id	Performance seat is linked to	INT	FK	NOT NULL	12 (from performance)
status	Status of seat	VARCHAR(25)		NOT NULL	AVAILABLE (default)
held_until	Time seat is held until	TIMESTAMP			1791088882

PERFORMANCE SEAT: one row is one seat that is available to reserve / purchase to a performance.

TICKET

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique ticket identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
performance_seat_id	Event seat identifier	INT	FK	NOT NULL (References PERFORMANCE_SEAT.id)	12 (from performance seating)
date_time	Time of ticket creation	TIMESTAMPTZ		NOT NULL	2026-10-22T22:39:55Z
price	Price of the ticket	NUMERIC(10,2)		NOT NULL	29.99

TICKET: one row is one ticket for a seat at a performance.

CART ITEM

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique cart item identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
cart_id	Cart identifier	INT	FK	NOT NULL	12 (from cart)
ticket_id	Ticket identifier	INT	FK	NOT NULL	12 (from ticket)

CART ITEM: one row is one item that sits inside a cart.

CART

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique cart identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
patron_id	Patron identifier to link cart to	INT	FK	NOT NULL	103 (from patron)
status	Status of cart	VARCHAR(50)		NOT NULL	ACTIVE (default)

CART: one row is one cart belonging to one patron.

ORDER

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique order identifier	INT	PK	NOT NULL UNIQUE	12 (generated)

cart_id	Cart identifier order is for	INT	FK	NOT NULL	12 (from cart)
date_time	Date and time of order	TIMESTAMPTZ		NOT NULL	2026-10-22T22:39:55Z
promo_code_id	Promo code applied to order	INT	FK	Nullable (References PROMO_CODE.id)	5 (from promo code)

ORDER: one row is one payment request for whatever items are in the cart.

PAYMENT

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique payment identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
order_id	Order identifier payment is for	INT	FK	NOT NULL, UNIQUE	12 (from order)
method	Method payment was made with	VARCHAR(50)		NOT NULL	MASTERCARD
amount	Amount paid	NUMERIC(10,2)		NOT NULL	56.99
date_time	Date and time of payment	TIMESTAMPTZ		NOT NULL	2026-10-22T22:39:55Z

PAYMENT: one row is one full payment for an order.

ROLE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique role identifier	INT, generated identity	PK	NOT NULL UNIQUE	1 (generated)
role	Name of the assigned role	VARCHAR(50)		NOT NULL, UNIQUE, Allowed values: 'ADMIN', 'STAFF', 'PATRON'	'PATRON'

ROLE: One row represents a specific operational role a staff member or patron has inside the organization.

PERFORMER

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique Person identifier	INT	PK/FK	NOT NULL UNIQUE (References PERSON.id)	103 (From person)
role	The performer's specific billing or marketing role	VARCHAR(50)			'Lead Actor'

PERFORMER: One row represents a performer account linked to a specific person for marketing purposes.

PRICE_TIER

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique price tier identifier	INT, generated identity	PK	NOT NULL UNIQUE	1 (generated)
section_id	Section identifier	INT	FK	Nullable (references SECTION.id)	NULL (default)
performance_id	Performance identifier	INT	FK	Nullable (references PERFORMANCE.id)	NULL (default)
stage_id	Stage identifier	INT	FK	Nullable (references STAGE.id)	12 (from stage)
event_id	Event identifier	INT	FK	Nullable (references EVENT.id)	12 (from event)
price	Base cost of the ticket	Numeric(10,2)		NOT NULL	29.99

PRICE_TIER: one row represents a specific pricing setting that can be applied dynamically to an event, performance, stage or section.

TICKET_SCAN

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique ticket scan identifier	INT, generated identity	PK	NOT NULL UNIQUE	1 (generated)
ticket_id	Identifier of the ticket scanned	INT	FK	NOT NULL (References TICKET.id)	12 (from ticket)
date_time	Date and time the scan happened	timestamp _{tz}		NOT NULL	2026-10-22T19:00:00Z
Success	Whether the scan lets you in	boolean		NOT NULL	TRUE

TICKET SCAN: one row is a single instance of a ticket being scanned for entry, taking into account potential rules for re-entry.

SECTION

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique section identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
stage_id	Stage the section belongs to	INT	FK	NOT NULL (References STAGE.id)	12 (from stage)

name	Name of the section	VARCHAR(50)		NOT NULL	'Balcony'
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SECTION: one row is a designated seating area in a stage.

PROMO_CODE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique promo code identifier	INT, generated identity	PK	NOT NULL UNIQUE	5 (generated)
code	Text code used by the patron	VARCHAR(50)		NOT NULL UNIQUE	'SAVE20'
d_off	Discount amount in dollars	NUMERIC(10,2)		Nullable	20.00
p_off	Discount amount in percentage	NUMERIC(5,2)		Nullable	0.15
active	Status of the promo code	BOOLEAN		NOT NULL	TRUE (default)
expiry_date	Date and time the code expires	TIMESTAMPTZ			2026-12-31T23:59:59Z

PROMO_CODE: one row is a discount code that can be applied to an order.

GENRE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique genre identifier	INT	PK	NOT NULL UNIQUE	8 (generated)
name	Name of the genre	VARCHAR(50)		NOT NULL	'Musical Theater'
active	Whether the genre is currently used or not	BOOLEAN		NOT NULL	TRUE (default)

GENRE: one row represents a category being used to classify events.

EVENT_GENRE

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
genre_id	Genre identifier	INT	PK/FK	NOT NULL (References GENRE.id)	8 (from genre)
event_id	Event identifier	INT	PK/FK	NOT NULL (References EVENT.id)	12 (from event)

EVENT_GENRE: One row is a link that connects a certain event to a specific genre.

PERFORMANCE_CAST

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
performer_id	performer identifier	INT	PK/FK	NOT NULL (References PERFORMER.id)	103 (from performer)
performance_id	Performance identifier	INT	PK/FK	NOT NULL (References PERFORMANCE.id)	12 (from performance)

PERFORMANCE_CAST: One row is a link that connects a performer or a specific performance.

EMAIL_LIST

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
person_id	person identifier	INT	PK/FK	NOT NULL (References PERSON.id)	103 (from person)
added	Date and time info was added to the list	TIMESTAMPZ		NOT NULL	2026-10-22T14:00:00Z

EMAIL_LIST: One row is person who has opted into marketing communications.

LOGIN_HISTORY

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique login record identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
person_id	Person attempting to log in	INT	FK	NOT NULL (References PERSON.id)	103 (from person)
ip_address	IP address of the login attempt	VARCHAR(50)		NOT NULL	'192.168.1.1'
success	Whether the login worked	BOOLEAN		NOT NULL	TRUE

LOGIN_HISTORY: one row is a record of a single log in attempt made by a user.

ERROR_LOG

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique error log identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
person_id	Person who triggered the error	INT	FK	Nullable (References PERSON.id)	103 (from person)

end_point	API or page endpoint	VARCHAR(255)		NOT NULL	'/api/checkout'
error_message	Short description of the error	VARCHAR(500)		NOT NULL	'Database timeout'
stack_trace	Full system stack trace	BYTEA			(Binary blob data)

ERROR_LOG: one row represents a system or application error that was generated during use.

PASSWORD_RESET_REQUEST

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique request identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
person_id	Person requesting the reset	INT	FK	NOT NULL (References PERSON.id)	103 (from person)
time	Date and time of request	TIMESTAMPTZ		NOT NULL	2026-10-22t10:05:00Z
success	Whether it was completed or not	BOOLEAN		NOT NULL	FALSE (default)
status	Current status of the request	VARCHAR(50)		NOT NULL	'PENDING'

PASSWORD_RESET_REQUEST: one row is a record kept of a user asking to recover their password to their account.

PERMISSION_CHANGE_LOG

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique log identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
person_id	Person whose permissions changed	INT	FK	NOT NULL (References PERSON.id)	103 (from person)
changed_by	Admin who made the change	INT	FK	NOT NULL (References PERSON.id)	12 (from person)
old_role	Previous role identifier	INT	FK	NOT NULL (References ROLE.id)	3 (from role)
new_role	New role identifier	INT	FK	NOT NULL (References ROLE.id)	1 (from role)
date_time	Date and time of change	TIMESTAMPTZ		NOT NULL	2026-10-22T08:30:00Z

PERMISSION_CHANGE_LOG: one row is an audit record kept of an account's security role being changed.

PERFORMANCE_STATUS_LOG

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique log identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
performance_id	Performance being tracked	INT	FK	NOT NULL (References PERFORMANCE.id)	12 (from performance)
changed_by	Staff who made the change	INT	FK	NOT NULL (References PERSON.id)	103 (from person)
previous	Old status	VARCHAR(50)		NOT NULL	'SCHEDULED'
new	New status	VARCHAR(50)		NOT NULL	'CANCELLED'
date_time	Date and time of change	TIMESTAMPZ		NOT NULL	2026-10-22T09:15:00Z

PERFORMANCE_STATUS_LOG: one row is a historical record kept of a performance's status changing.

AUDIT_LOG

Attribute	Description	Data Type	Key	Validation Rules	Sample / Default
id	Unique audit log identifier	INT	PK	NOT NULL UNIQUE	12 (generated)
staff_id	Staff member performing the action	INT	FK	NOT NULL (References STAFF.id)	103 (from staff)
table_name	Database table affected	VARCHAR(50)		NOT NULL	'EVENT'
record_id	PK of the affected record	INT		NOT NULL	12
action	Type of action done	VARCHAR(50)		NOT NULL	'UPDATE'
old_value	Data before the change	BYTEA			(Binary blob data)
new_value	Data after the change	BYTEA			(Binary blob data)
changed_at	Date and time of action	TIMESTAMPZ		NOT NULL	2026-10-22T11:45:00Z
ip_address	IP address of the staff member	VARCHAR(45)		NOT NULL	'192.168.1.1'

AUDIT_LOG: one row is security record keeping track and detailing any modifications made to data by staff members.